

A Sample Physics Document

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Abstract

This is a sample LaTeX document containing mathematical equations, tables, matrices, figures, lists, and references. It can be used to test conversion from $\LaTeX$ to HTML.

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1 Introduction

Classical mechanics provides a powerful framework for describing physical systems. The motion of a system can be obtained from the Lagrangian

$$L = T - V, \tag{1}$$

where T is the kinetic energy and V is the potential energy. The Euler–Lagrange equation is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0. \tag{2}$$

Equation (equation2) determines the equations of motion of the system.

2 Mathematical Expressions

Consider a simple harmonic oscillator with Lagrangian

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2. \quad (3)$$

Using the Euler–Lagrange equation,

$$\frac{\partial L}{\partial x} = -kx, \quad (4)$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}. \quad (5)$$

Therefore,

$$m\ddot{x} + kx = 0. \quad (6)$$

Defining

$$\omega = \sqrt{\frac{k}{m}}, \quad (7)$$

we obtain

$$\boxed{\ddot{x} + \omega^2 x = 0}. \quad (8)$$

The general solution is

$$x(t) = A\cos(\omega t) + B\sin(\omega t). \quad (9)$$

2.1 A More Complicated Equation

Consider the integral

$$I = \int_0^\infty \frac{\ln x}{1+x^3} dx. \quad (10)$$

A useful identity is

$$\int_0^\infty \frac{x^{a-1}}{1+x^b} dx = \frac{\pi}{b} \csc\left(\frac{\pi a}{b}\right), \quad 0 < a < b. \quad (11)$$

Differentiating with respect to a gives

$$\frac{d}{da} \int_0^\infty \frac{x^{a-1}}{1+x^b} dx = \int_0^\infty \frac{x^{a-1} \ln x}{1+x^b} dx. \quad (12)$$

3 Matrices

A general 2×2 matrix can be written as

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (13)$$

Its determinant is

$$\det A = ad - bc. \quad (14)$$

The inverse, provided $\det A \neq 0$, is

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \quad (15)$$

For angular momentum, the commutation relation is

$$[J_i, J_j] = i\hbar \epsilon_{ijk} J_k. \quad (16)$$

For spin- $\frac{1}{2}$, the generators are

$$J_i = \frac{\hbar}{2} \sigma_i, \quad (17)$$

where the Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (18)$$

4 Tables

Table 1 shows some example particle properties.

Table 1: Properties of selected particles.			
Particle	Mass (kg)	Charge (e)	Spin
Electron	9.11×10^{-31}	-1	$\frac{1}{2}$
Proton	1.67×10^{-27}	+1	$\frac{1}{2}$
Neutron	1.67×10^{-27}	0	$\frac{1}{2}$
Photon	0	0	1

5 Lists

Important concepts in classical mechanics include:

- Configuration space
- Generalized coordinates

- Lagrangian mechanics
- Hamiltonian mechanics
- Canonical transformations

The usual procedure for solving a Lagrangian problem is:

1. Choose generalized coordinates.
2. Write the kinetic energy.
3. Write the potential energy.
4. Construct the Lagrangian.
5. Apply the Euler–Lagrange equations.

6 Figures

Figure figure1 is an example figure.

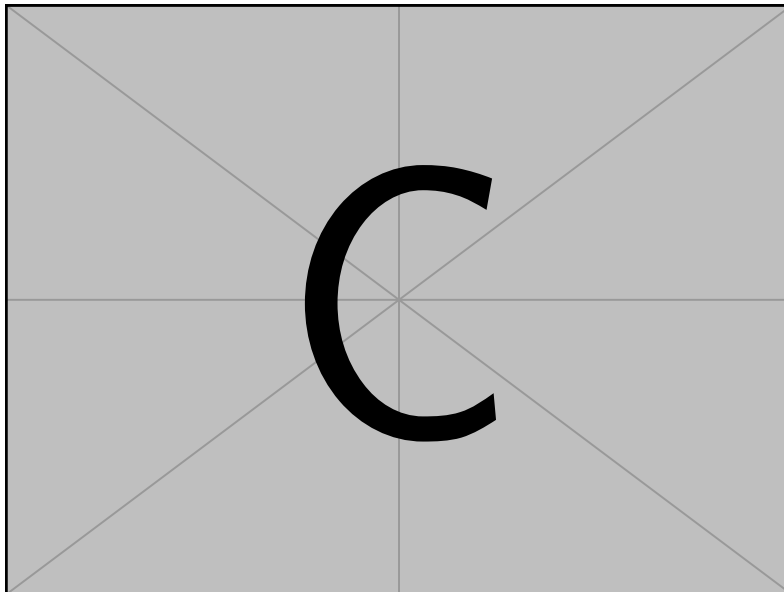


Figure 1: An example figure.

7 Physics Example

Consider a simple pendulum of length L and mass m .

The position of the bob can be written as

$$x = L \sin \theta, \quad y = -L \cos \theta. \quad (19)$$

The velocity is

$$v^2 = L^2 \dot{\theta}^2. \quad (20)$$

Therefore, the kinetic and potential energies are

$$T = \frac{1}{2} m L^2 \dot{\theta}^2, \quad (21)$$

$$V = m g L (1 - \cos \theta). \quad (22)$$

Hence,

$$L = \frac{1}{2} m L^2 \dot{\theta}^2 - m g L (1 - \cos \theta). \quad (23)$$

Applying the Euler–Lagrange equation gives

$$m L^2 \ddot{\theta} + m g L \sin \theta = 0. \quad (24)$$

Thus,

$$\boxed{\ddot{\theta} + \frac{g}{L} \sin \theta = 0}. \quad (25)$$

For small oscillations,

$$\sin \theta \approx \theta, \quad (26)$$

and therefore

$$\ddot{\theta} + \frac{g}{L} \theta = 0. \quad (27)$$

The period is

$$T = 2\pi \sqrt{\frac{L}{g}}. \quad (28)$$

8 References

For more information, see standard texts on classical mechanics such as [1].

References

- [1] H. Goldstein, C. Poole, and J. Safko, *Classical Mechanics*, 3rd ed., Pearson, 2002.
- [2] D. J. Griffiths, *Introduction to Quantum Mechanics*, 3rd ed., Cambridge University Press, 2018.

9 Conclusion

This document contains:

- Mathematical equations
- Aligned equations
- Matrices
- Tables
- Figures
- Lists
- Cross-references
- Citations
- Sections and subsections

These features make it useful for testing a $\text{\LaTeX}$ to HTML conversion workflow.